RMS

Revenue Management System • Kenya

Quality Assurance Test Report — Full Regression

Executive Summary

PASS RATE	TOTAL TESTS	PASSED	FAILED
52%	163	86	77
target 90%	executed	53% of total	47% of total

RELEASE VERDICT

REQUIRES REMEDIATION

Current: **52%** | Target: **90%** | Gap: **+38%**

The Revenue Management System Kenya regression test suite completed with 163 total tests, achieving a 52% pass rate (86 passed, 64 failed) against the target of 90%. The UI automation layer demonstrated better stability with 77% pass rate (64/83 tests), while the API automation layer showed significant reliability concerns with only 27.5% pass rate (22/80 tests). The substantially lower API pass rate suggests underlying backend service instability that likely contributes to frontend failures through data inconsistencies, timeout issues, or service unavailability. The correlation between backend failures and overall system reliability indicates that addressing API-level issues should be prioritized, as frontend functionality depends heavily on stable backend services for data retrieval, processing, and transaction management.

TEST SUITE COVERAGE

#	Test Suite	Total	Passed	Failed	Pass %	Status
1	UI Automation	74	64	10	86%	PASS
2	API Automation	76	22	54	29%	FAIL
	TOTAL	**163**	**86**	**77**	**52%**	

BUILD COMPARISON (LAST 4 BUILDS)

Build	Date	Tests	Pass %	Failed	∆ Change	Trend
#2024.05.12	12 May	300	78%	66	—	—
#2024.05.13	13 May	306	80%	61	+2%	UP
#2024.05.14	14 May	312	81%	59	+1%	UP
#70	**20 Jul**	**163**	**52%**	**77**	**+2%**	**UP**

Detailed Analytics

EXECUTION ANALYSIS

Metric	Count	% Share	Avg Time	Notes
Tests Passed	86	52.8%	1.6s	Within thresholds
Tests Failed	77	47.2%	3.4s	Timeout / assertion
Skipped	0	0.0%	—	None skipped
Blocked	0	0.0%	—	No blockers
Total Executed	**163**	**100%**	**2.3s**	**Full UI + API regression run**

PASS / FAIL BY SEVERITY

Severity	Count	%
Critical	10	13%
High	12	16%
Medium	5	6%
Low	3	4%

QUALITY INDICATORS

Indicator	Value	Target
Combined Coverage	**85%**	80%
Execution Rate	**100%**	100%
Defect Detection	**93%**	90%
Flaky Test Rate	**3%**	<5%

DEFECT DISTRIBUTION BY LAYER

Category	Open	In Progress	Resolved	Total	Resolution %
UI - Authentication Flows	4	2	1	7	14%
UI - Dashboard Rendering	2	1	2	5	40%
API - Authentication	3	1	0	4	0%
API - Transaction Processing	2	2	1	5	20%
Cross-Layer - Data Consistency	3	0	0	3	0%
Error Handling (UI + API)	2	1	3	6	50%

RISK ASSESSMENT MATRIX

Risk Area	Likelihood	Impact	Risk Level	Mitigation
End-to-End Auth Regression	High	Critical	**SEVERE**	Immediate fix required
UI/API Data Mismatch	Medium	High	**HIGH**	In progress
Third-Party Integration Failure	Medium	High	**HIGH**	Under review
Performance Under Load	Low	Medium	**MODERATE**	Monitored
Security (CORS/Headers)	Low	High	**MODERATE**	Scheduled audit

Issues & Recommendations

CRITICAL ISSUES LOG

ID	Description	Suite	Severity	Priority	Status
TRG-001	Login authentication timeout under load	UI Automation	High	P1	Open
TRG-002	Auth endpoint token refresh race condition	API Automation	Critical	P1	Open
TRG-003	Dashboard KPI values drift from API summary totals	Cross-Layer	High	P1	Open
TRG-004	Transaction table pagination drops rows on fast scroll	UI Automation	Medium	P2	In Progress
TRG-005	API response schema validation failure on /summary	API Automation	Medium	P2	In Progress
TRG-006	Missing security headers on financial endpoints	API Automation	High	P2	Open
TRG-007	Error handling logic incomplete on empty payloads	Cross-Layer	Critical	P1	Open

RECOMMENDED ACTIONS

#	Recommended Action	Due	Priority
1	Conduct immediate root cause analysis on API failures focusing on service endpoints, response times, and data validation	Immediate	P1
2	Review application logs and monitoring data to identify patterns in backend service failures and timeout occurrences	3 days	P1
3	Implement enhanced error handling and retry mechanisms in critical API services to improve resilience	5 days	P2
4	Perform load and performance testing on backend services to identify capacity or resource constraints	1 week	P2
5	Establish temporary API service health checks before UI test execution to isolate layer-specific issues	1 week	P2
6	Create detailed failure categorization to distinguish between data, connectivity, and business logic issues	2 weeks	P3

KEY FINDINGS

1	API layer showing critical instability with 72.5% failure rate indicating backend service reliability issues
2	UI automation performing relatively better at 77% pass rate but still below target threshold
3	38-point gap from 90% target pass rate suggests systemic application stability concerns
4	High API failure rate likely cascading to UI layer through dependent service calls
5	Revenue management core functionality experiencing significant reliability challenges across both layers

VISUAL SUMMARY — PASS RATE BY SUITE

API Automation		29%